

Ryan Tomich

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Education

Massachusetts Institute of Technology

Sep 2023 - May 2027

Bachelor of Science in Computer Science and Engineering; GPA: 4.8/5.0

Cambridge, MA

Relevant Courses: ML with Light, Engineering Design and Rapid Prototyping, Software Construction, Computation Structures, Design and Analysis Algorithms, Intro to Machine Learning, Linear Algebra, Introduction to Probability, Digital Systems Lab

Skills

Programming Languages: Python, C, C++, TypeScript & Node.js, RISC-V Assembly, Verilog HDL

Deep Learning & AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Contrastive Learning, embeddings, semantic deduplication, PyTorch, Hugging Face, Transformers, ONNX, TVM

Compilers & Systems: Compiler IR, Multi-Target Compilation, Hardware-Software Co-Design, Hardware Profiling & Benchmarking

Developer Tools: Git, Conda, VSCode, LaTeX, NumPy, Matplotlib, Pytest, black, SSH, Slurm, tmux

Professional Experience

Jump Trading

Jun 2026 -

Software Engineer Intern

Chicago, IL

MIT-IBM Watson AI Lab

Jun 2025 - Present

AI Research Intern

Cambridge, MA

- Developed a data-augmentation framework that expanded knowledge-base coverage and enhanced quality; implemented semantic deduplication to shrink size by 50% with no loss in generation quality.
- Built a RAG-based multi-agent pipeline using Qwen3-Embedding, GraphCodeBERT, and CLIP for code optimization, improving LLM-generated code performance by 1.4 $\times$.
- Achieved a 30% accuracy boost over traditional k-NN baselines by benchmarking and optimizing code-text retrieval methods and applying statistical evaluation as an inexpensive quality heuristic.

MIT EECS

Feb 2025 - May 2025

Fundamentals of Programming Lab Assistant

Cambridge, MA

- Held office-hour and code-review sessions for 300 MIT undergraduates, clarifying Python, data-structure, and algorithm concepts.
- Pre-tested 13 programming labs and companion readings before release to sharpen skills and raise content quality.

MIT Research Laboratory of Electronics: Quantum Photonics & AI Group

Feb 2024 - Feb 2025

Undergraduate Researcher

Cambridge, MA

- Developed *RyanTomich/LightCode*, a compiler framework for LLM inference on photonic-electronic systems.
- Built an arithmetic hardware simulator and optimizing compiler to model latency and energy for GPUs and integrated photonic accelerators.
- Designed a novel IR for mapping operations to optimal hardware, reducing simulated energy by 50% via subgraph selection, graph partitioning, and shortest-path algorithms that reduces simulated energy consumption by 50%.
- Created *RyanTomich/NP_GPT2*, a ground-up NumPy implementation of GPT-2 and the transformer architecture.

Town Pump

Jun 2023 - Aug 2023

Information Technology Technician Intern

Butte MT

- Facilitated a state-wide Windows 11 migration and phone system integration by configuring 73 computers and 100 phones across 23 stores.

Leadership and Project Experience

MIT Motorsports

May 2023 - Present

Controls Software Engineer

Cambridge, MA

- Co-authored *DesignQA*, an LLM-powered design query and CAD database benchmark, published in ASME Journal of Computing and Information Science in Engineering.
- Integrated control algorithms into IPGCar digital twin, enabling pre-assembly testing, validation, and driver training.
- Developed torque vectoring and traction control algorithms for a 4WD electric race car improving corner exit speed by 7%.
- Designed launch control strategies by modeling tire slip dynamics to enhance acceleration and cornering performance.
- Created motor efficiency maps from torque-speed data for power-limiting compliance; contributed to 1st place in Design at Formula Hybrid 2024 with a perfect controls score.

Leadership Training & Development

- *Undergraduate Practice Opportunities Program(UPOP)*: completed year-long program on leadership and professional-development.
- *Gordon-MIT Engineering Leadership Program (GEL)*: Year-long program on Engineering leadership.
- *Policy Debate*: Over 50 hours of competition speaking experience, Cut over 1200 pieces of evidence, Wrote 15 technical cases, contributed to 23 briefs. 2 Nationals appearances.